MetaPilot User Manual

Kai Cheng, Ph.D.

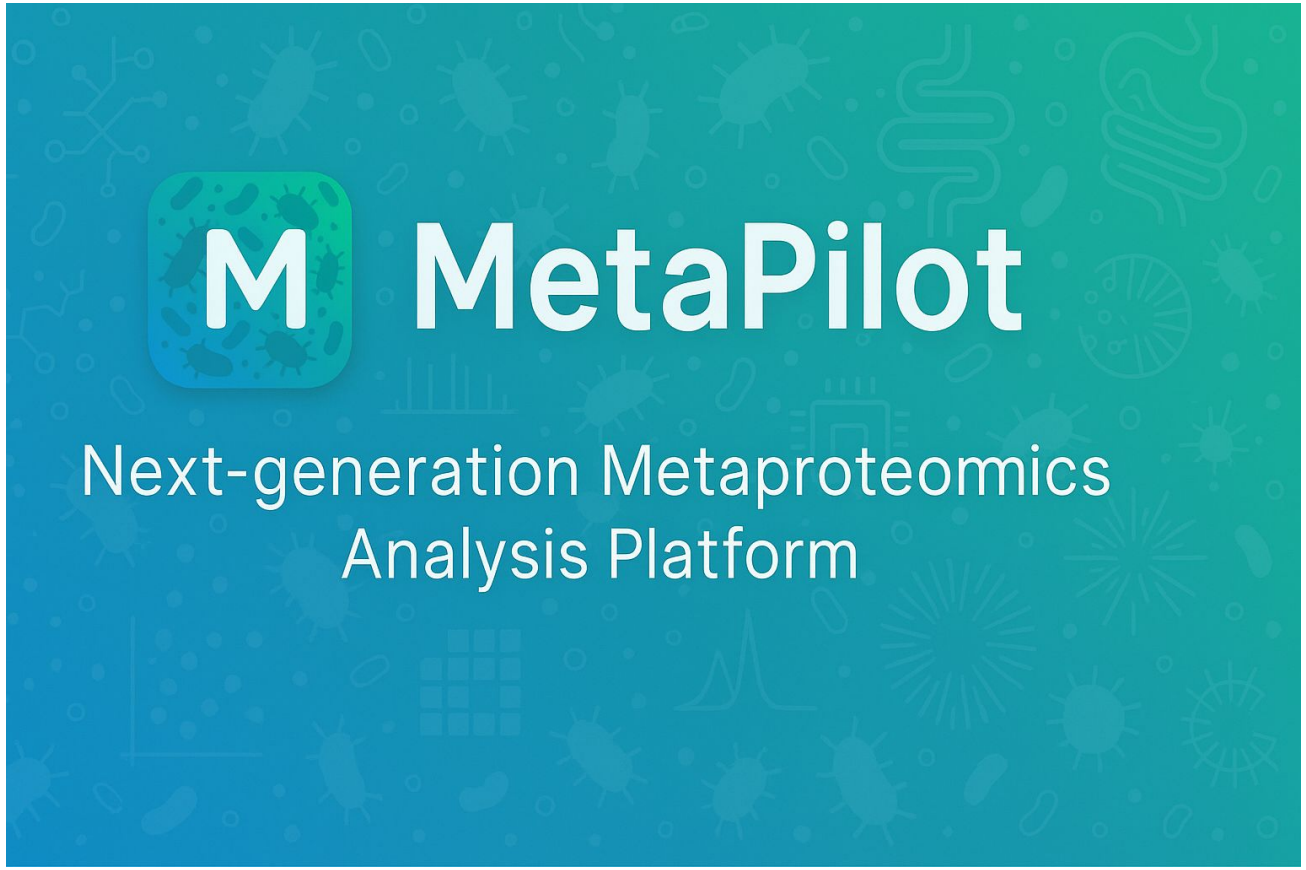
AI & Metaproteomics for Human Gut Health group

Quadram Institute, United Kingdom



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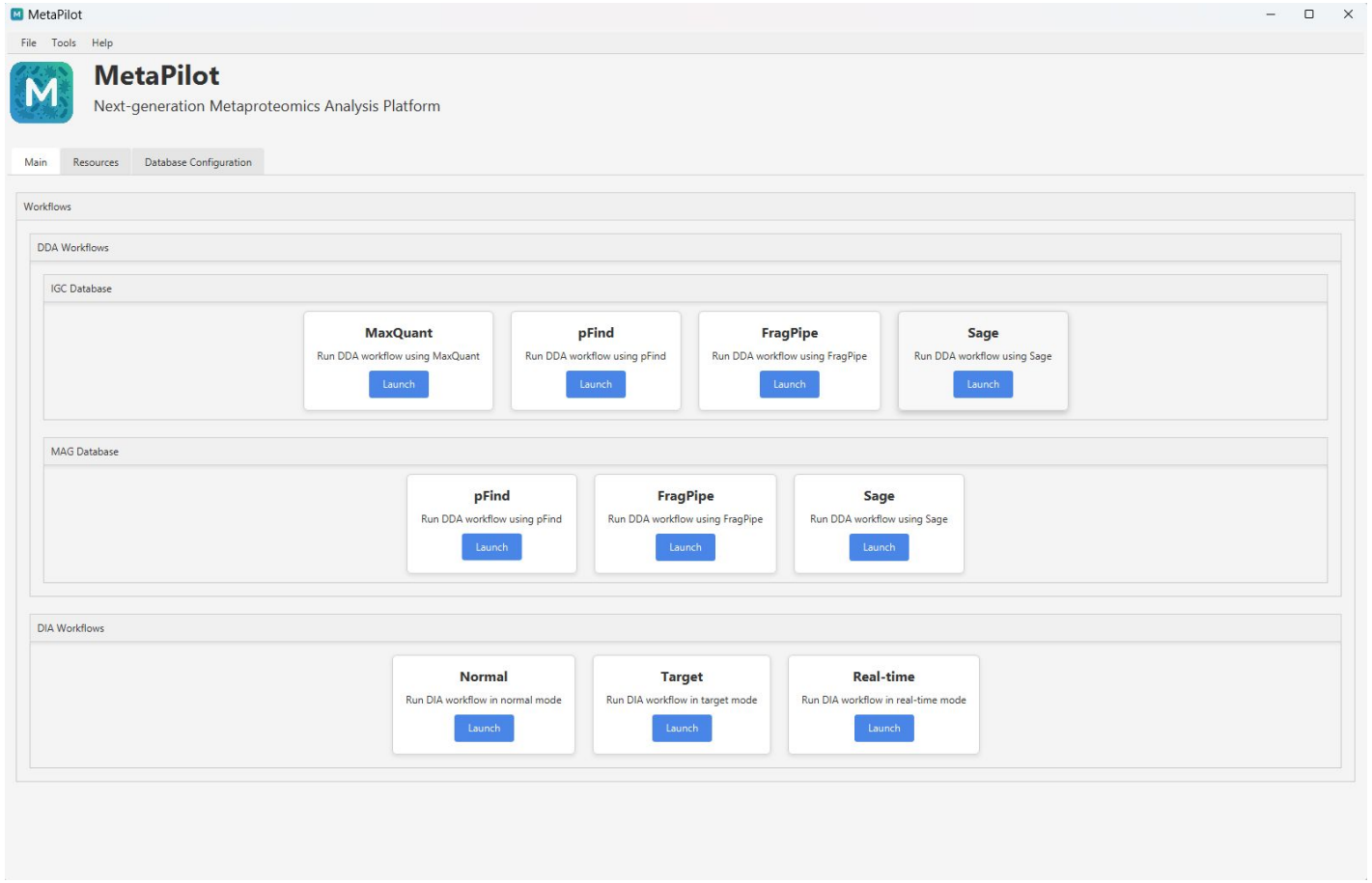
Introduction

MetaPilot integrates multiple search engines and workflows within a unified JavaFX GUI.

Supports DDA and DIA workflows for metaproteomics analysis.

Key Features:

- Multi-workflow support: DDA (IGC & MAG) and DIA.
- Multi-engine integration: MaxQuant, pFind, FragPipe, Sage, DIA-NN.
- Centralized tool and database management.



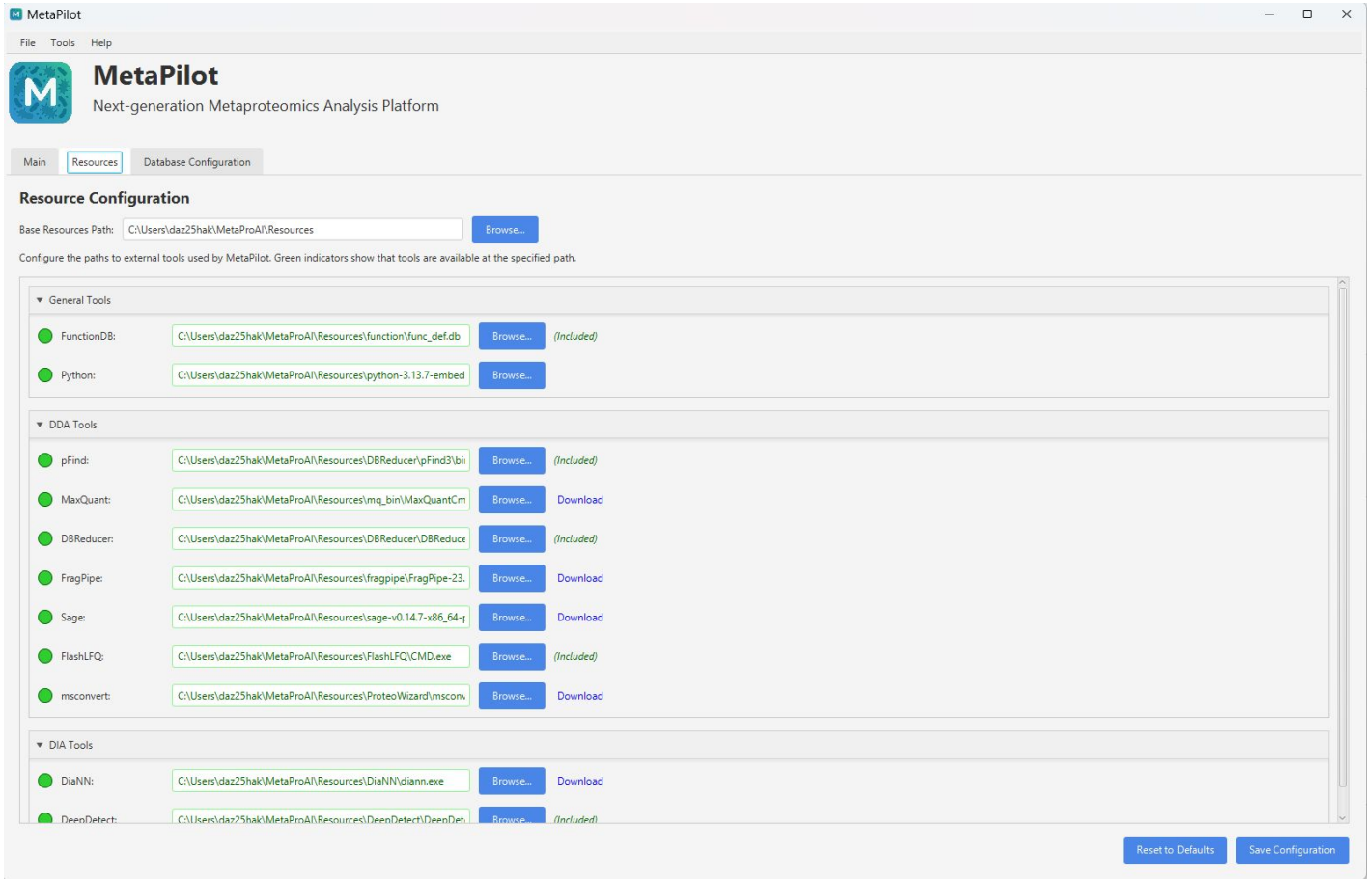
System Requirements

Environment required:

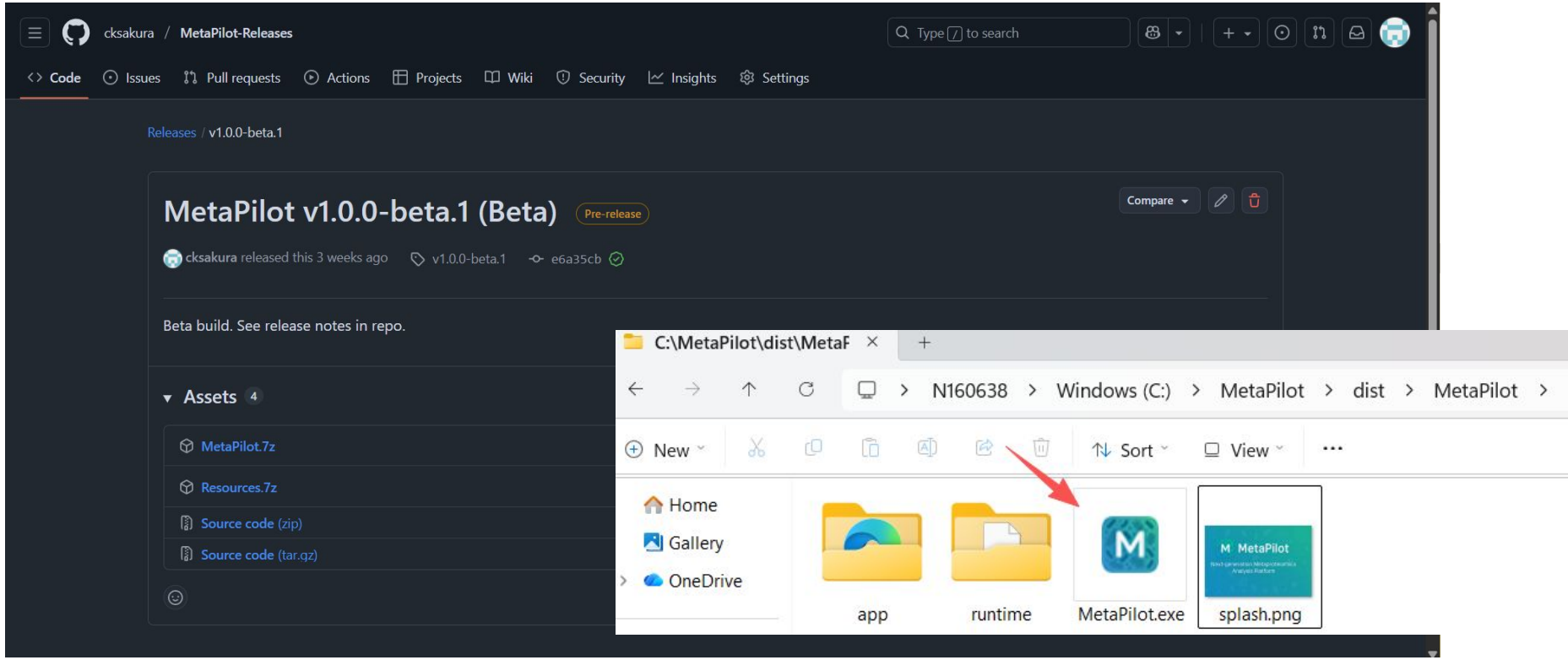
- OS: Windows
- Hardware: minimum 8GB+ RAM (higher RAM enables the processing of higher number of parallel tasks), 10GB free space, multi-core CPU

External Tools Required:

- MaxQuant, pFind, FragPipe, Sage, DIA-NN
- msconvert (ProteoWizard), FlashLFQ, Python, FunctionDB



Installation & Setup



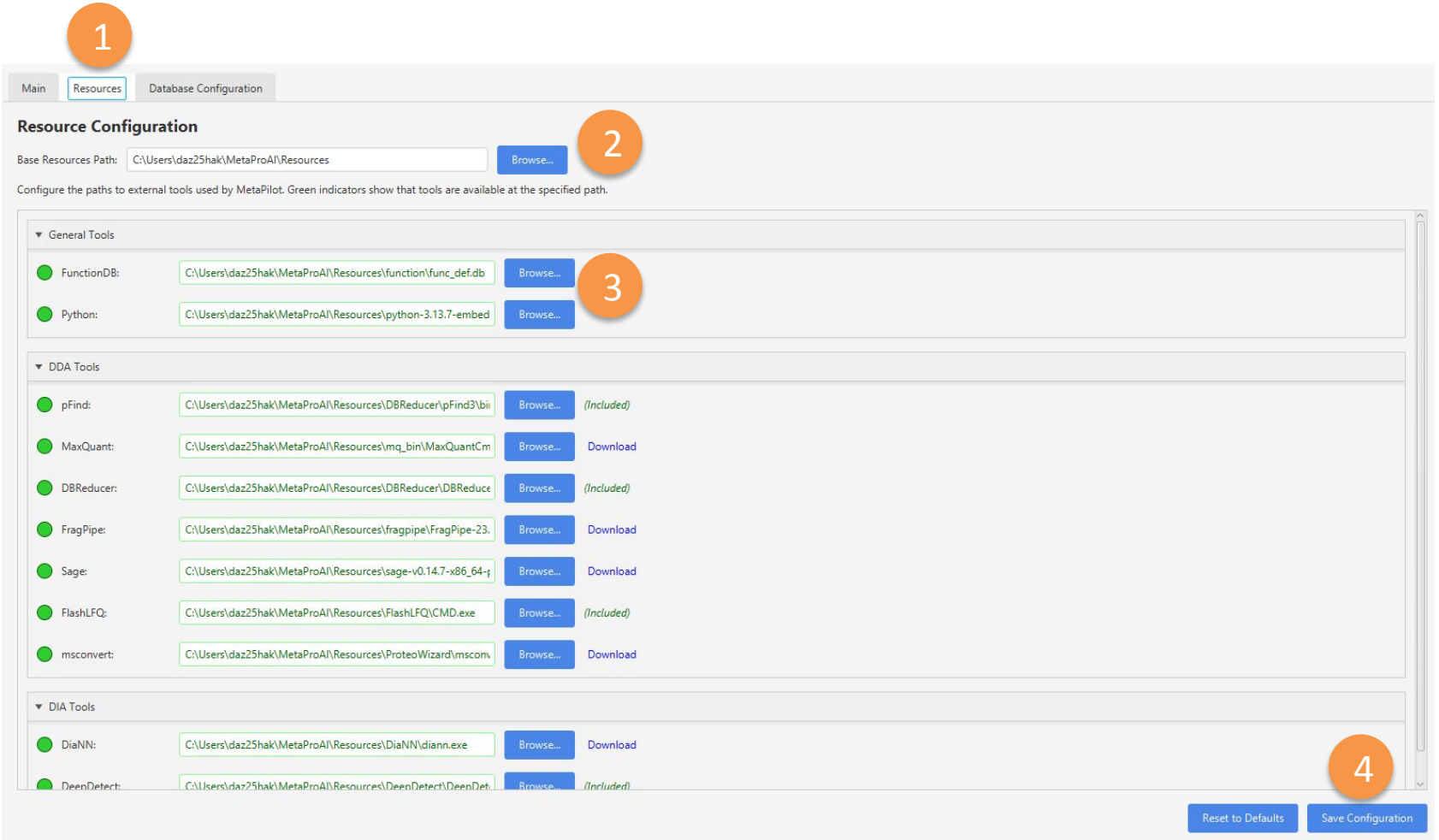
- Download the zip file, extract it and run the MetaPilot.exe directly.
- Download the Resources.7z and decompress the files.

Resource Configuration

Centralized management of external tool paths in config/meta_resources.params, configure via GUI or edit manually.

Steps (GUI):

1. Open 'Resources' tab.
2. Set base resource path (the extracted folder of the “Resources.7z”).
3. Tools not provided from the “Resources.7z” could be downloaded by the click the “Download”.
4. Save configuration.

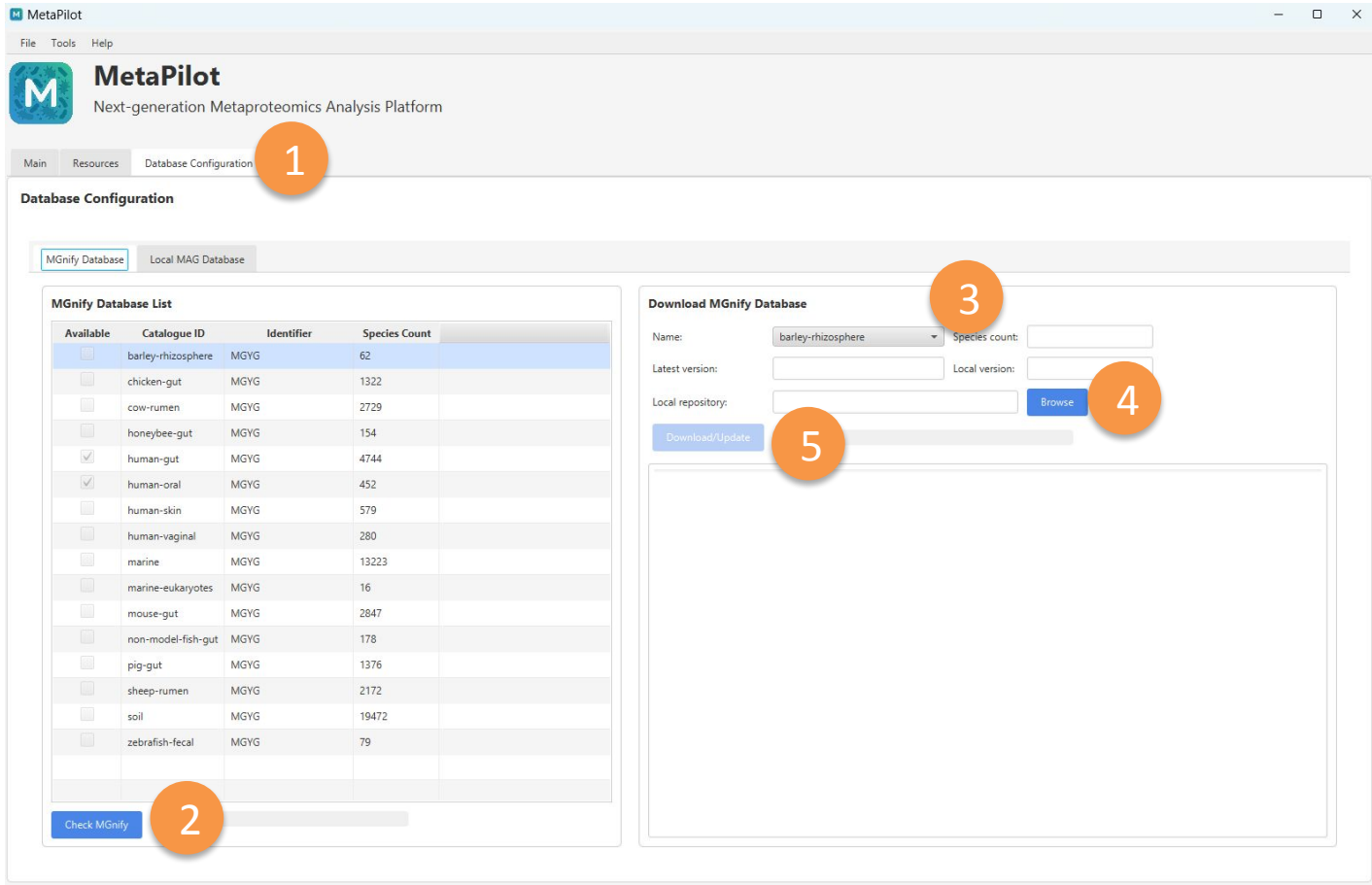


MAG Database Setup

Configure MAG databases via Database Configuration tab or JSON file, supports MGnify and local custom databases.

Steps (GUI):

1. Open “Database Configuration” tab.
2. Click “Check Mgnify”.
3. Select database.
4. Set the local path.
3. Click 'Download' to fetch database files.

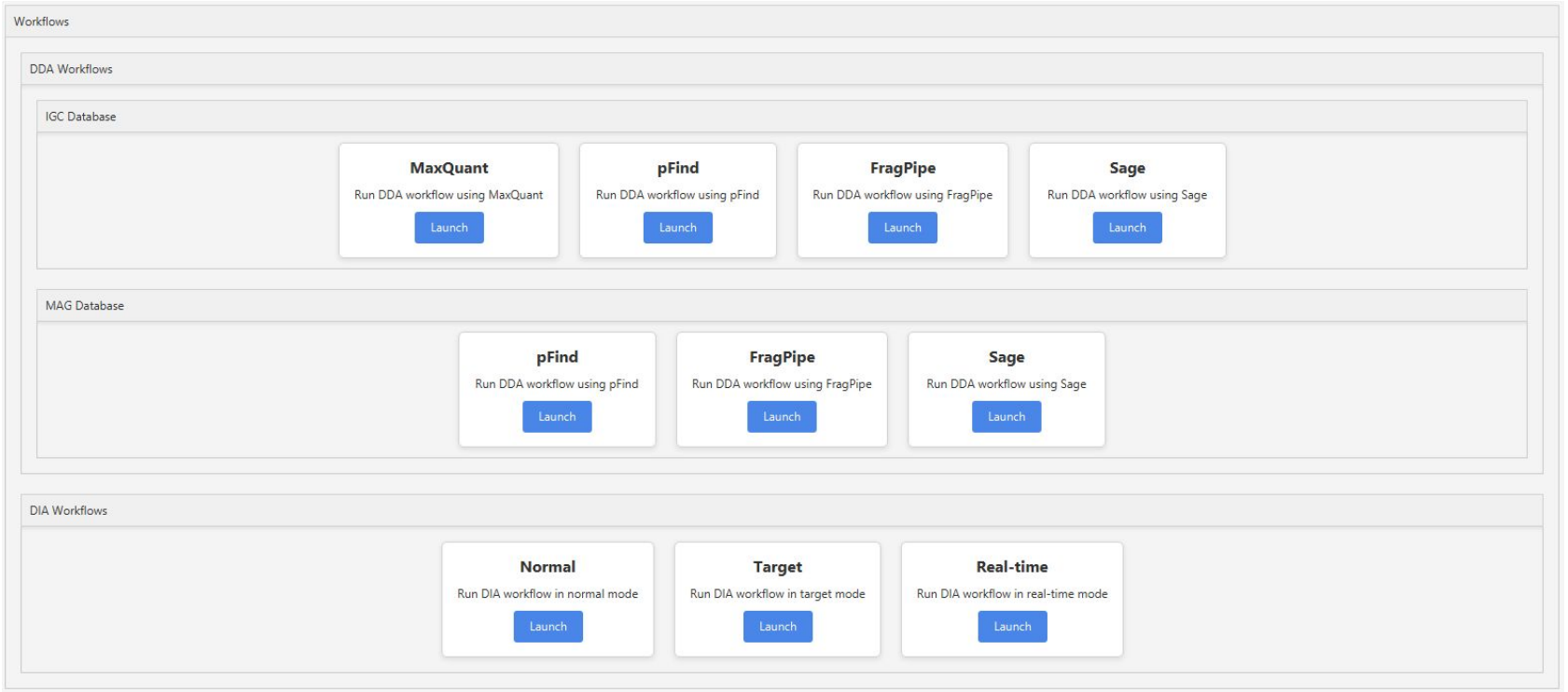


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Workflow Configuration

- MetaPilot supports both DDA and DIA workflows.
- DDA Workflows: IGC & MAG with engines (MaxQuant, pFind, FragPipe, Sage).
- DIA Workflows: Normal, Target, Real-time modes.

- Common Parameters: Input/Output, enzyme, modifications.
- Advanced: threads, sample-specific DB, quantification settings.

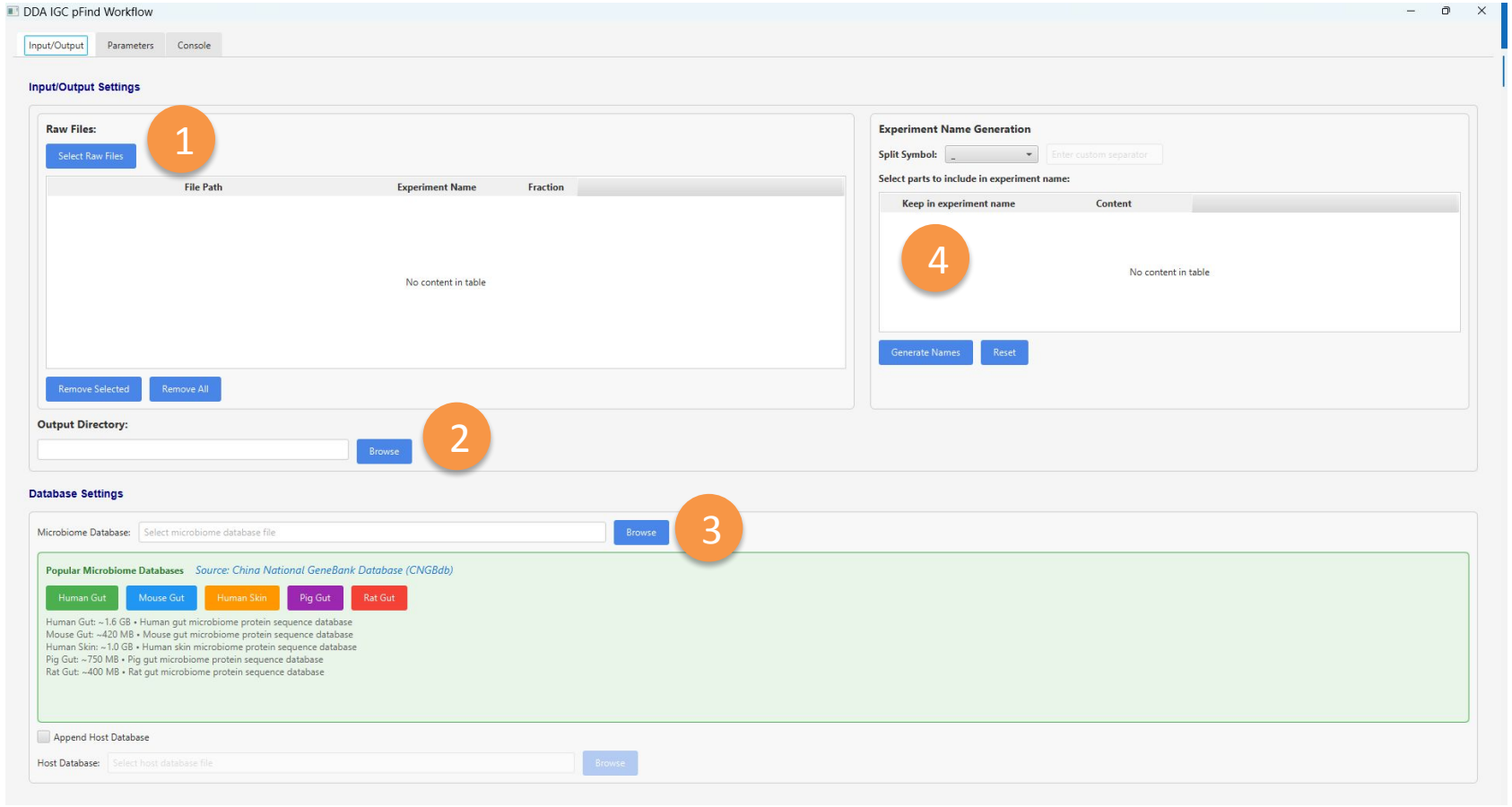


Running Analyses: DDA IGC Workflows

DDA IGC workflow: searching against an integrated genome catalog database.

Steps (GUI):

1. Input raw files.
2. Set output directory,.
3. Define databases.
4. Set the experiment names (optional).



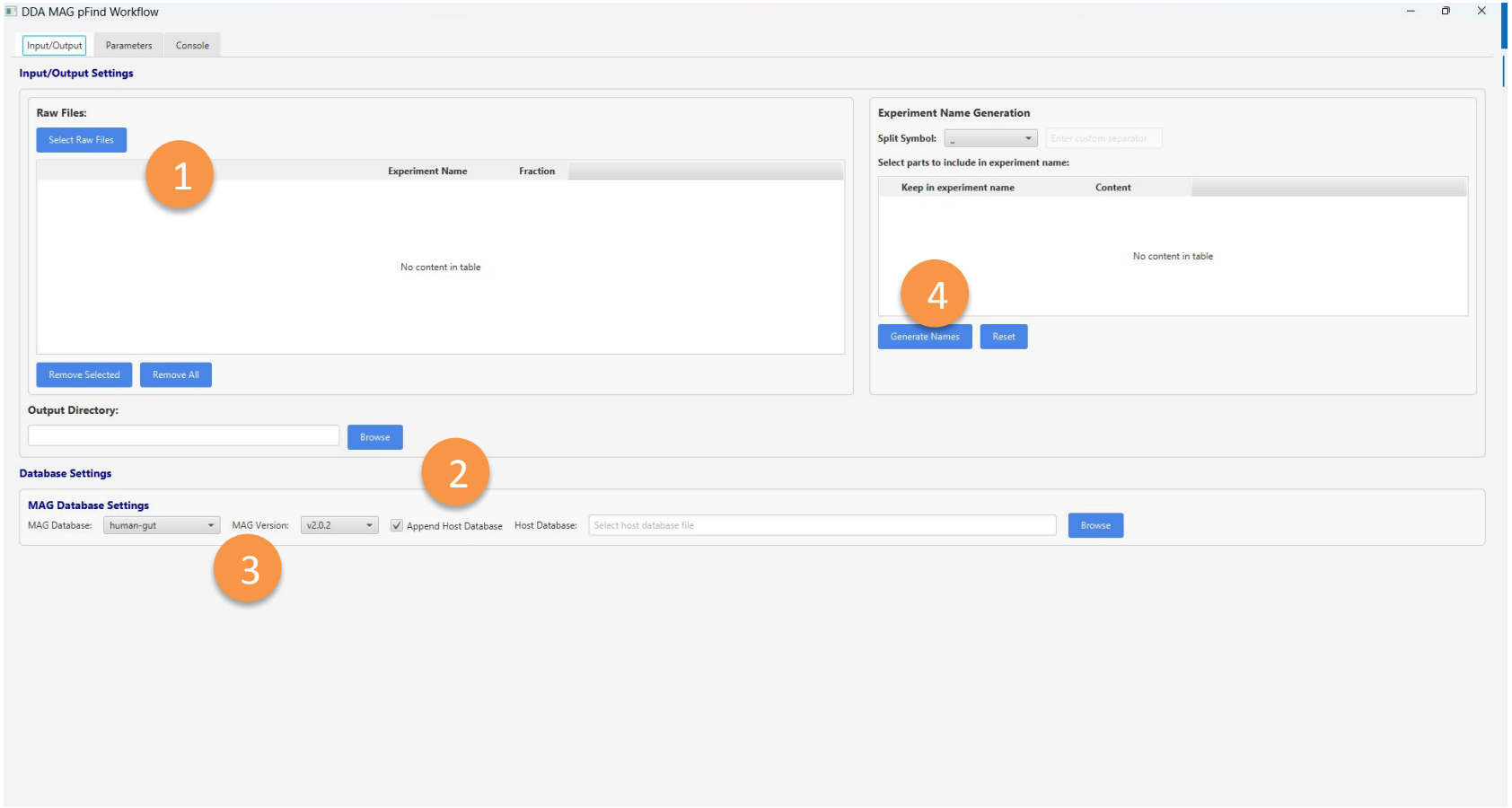
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Running Analyses: DDA MAG Workflows

DDA MAG workflow: searching against a metagenome-assembled genomes (MAG) database.

Steps (GUI):

1. Input raw files.
2. Set output directory,.
3. Select the MAG database (download it first in the MAG Database Setup).
4. Set the experiment names (optional).



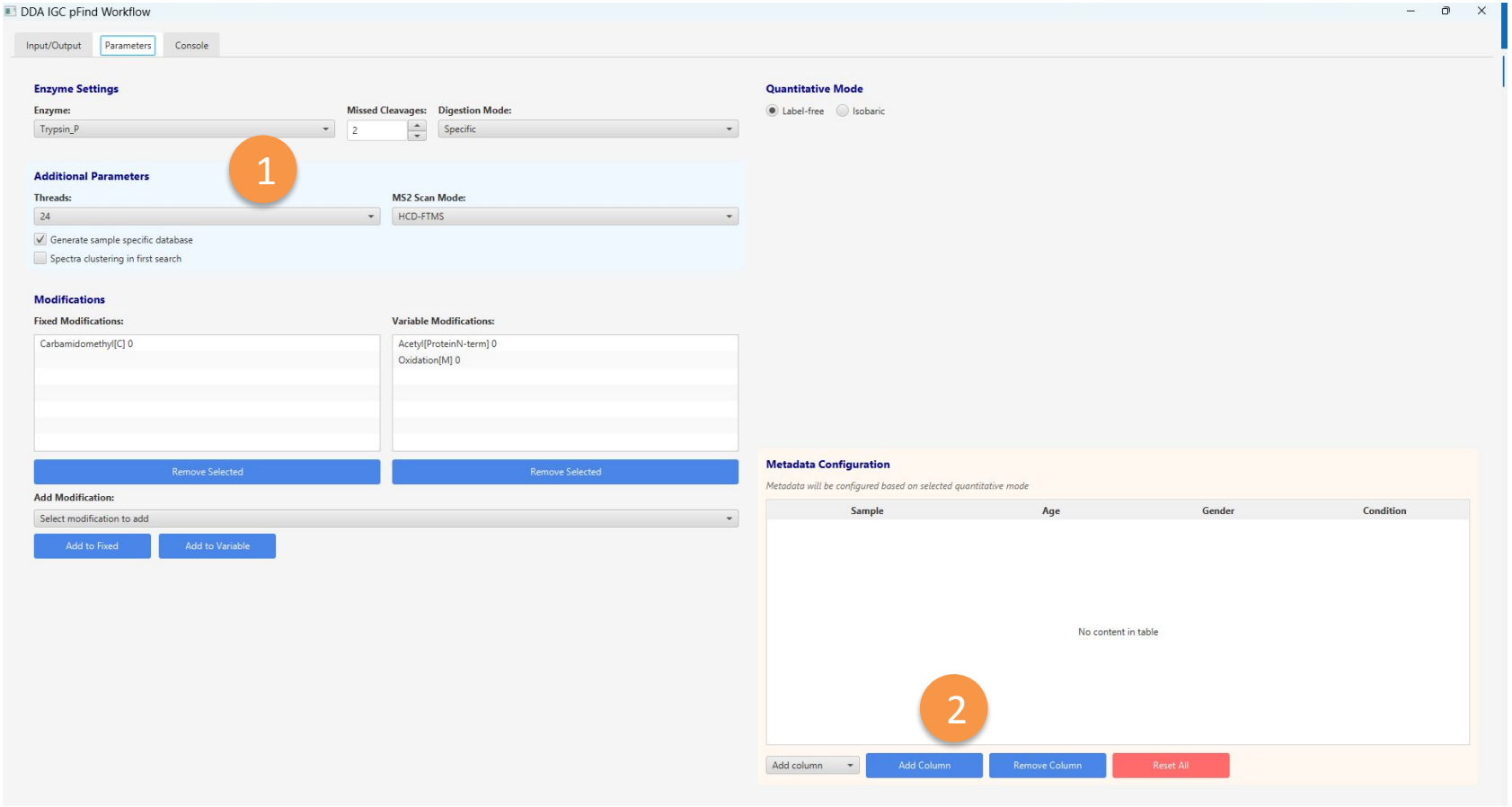
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Running Analyses: set database search parameters

Steps (GUI):

1. configure the common parameters used in proteomics database search.

2. Optional: set the Metadata.



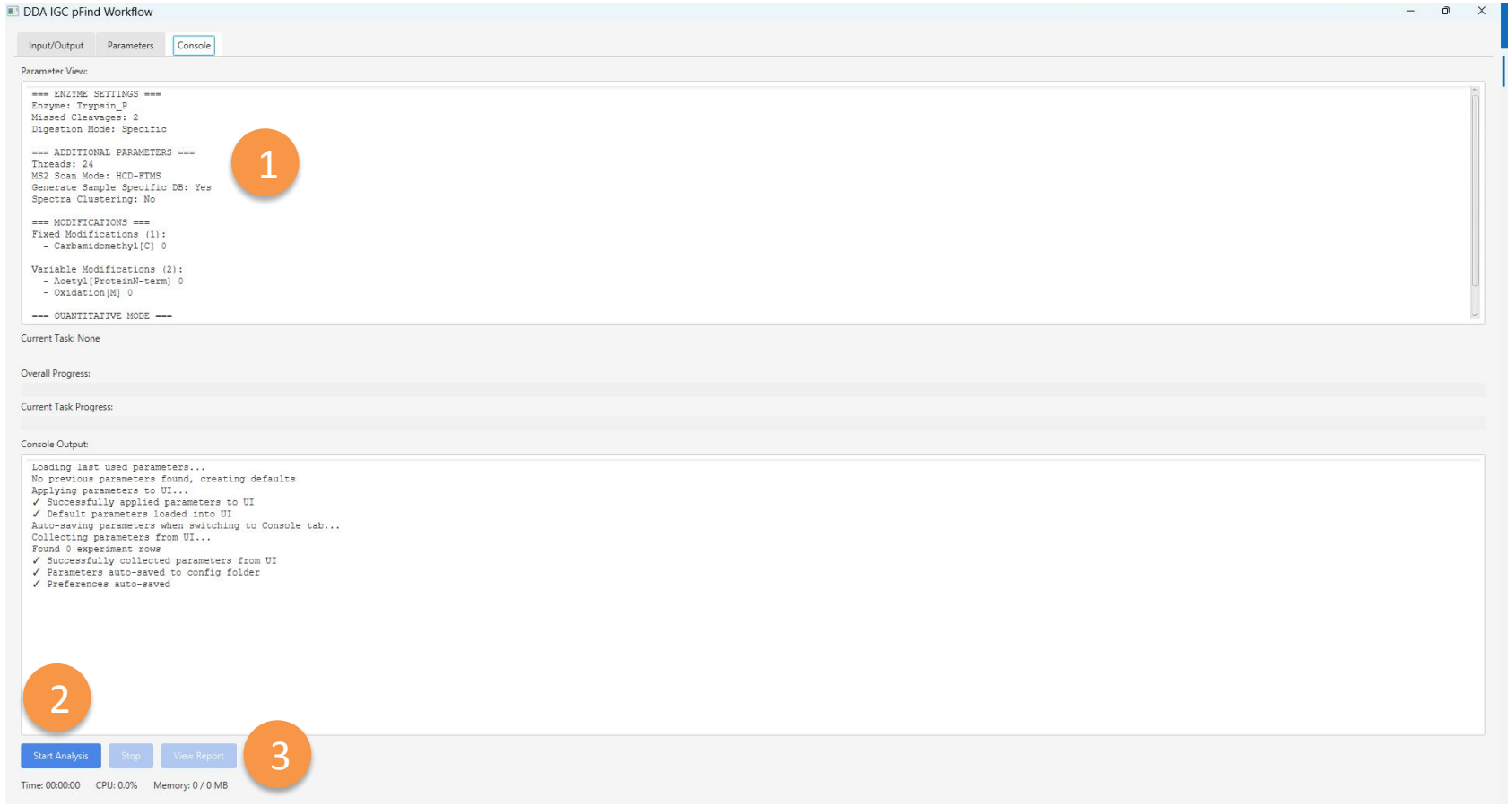
Running Analyses: start analysis

Steps (GUI):

1. Check the configured parameters.

2. Start analysis and monitor progress in console.

3. Results saved to output directory → View Report.

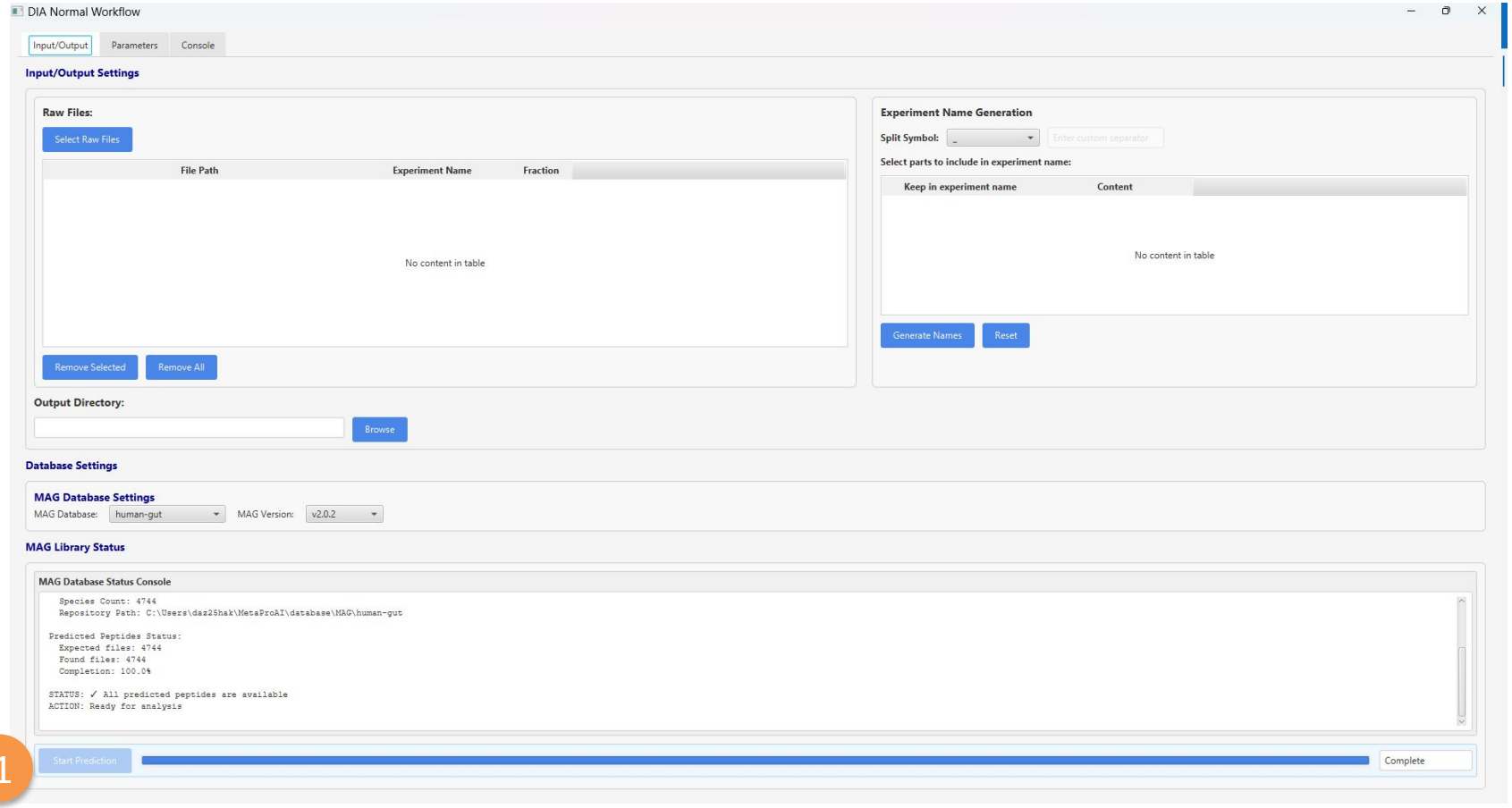


Running Analyses: DIA Workflows

MetaPilot offers advanced DIA workflows tailored for metaproteomics.

Step (GUI):

1. The DIA workflows requires a in-silico digested peptide sequence repository, if it is not available, generate it first.

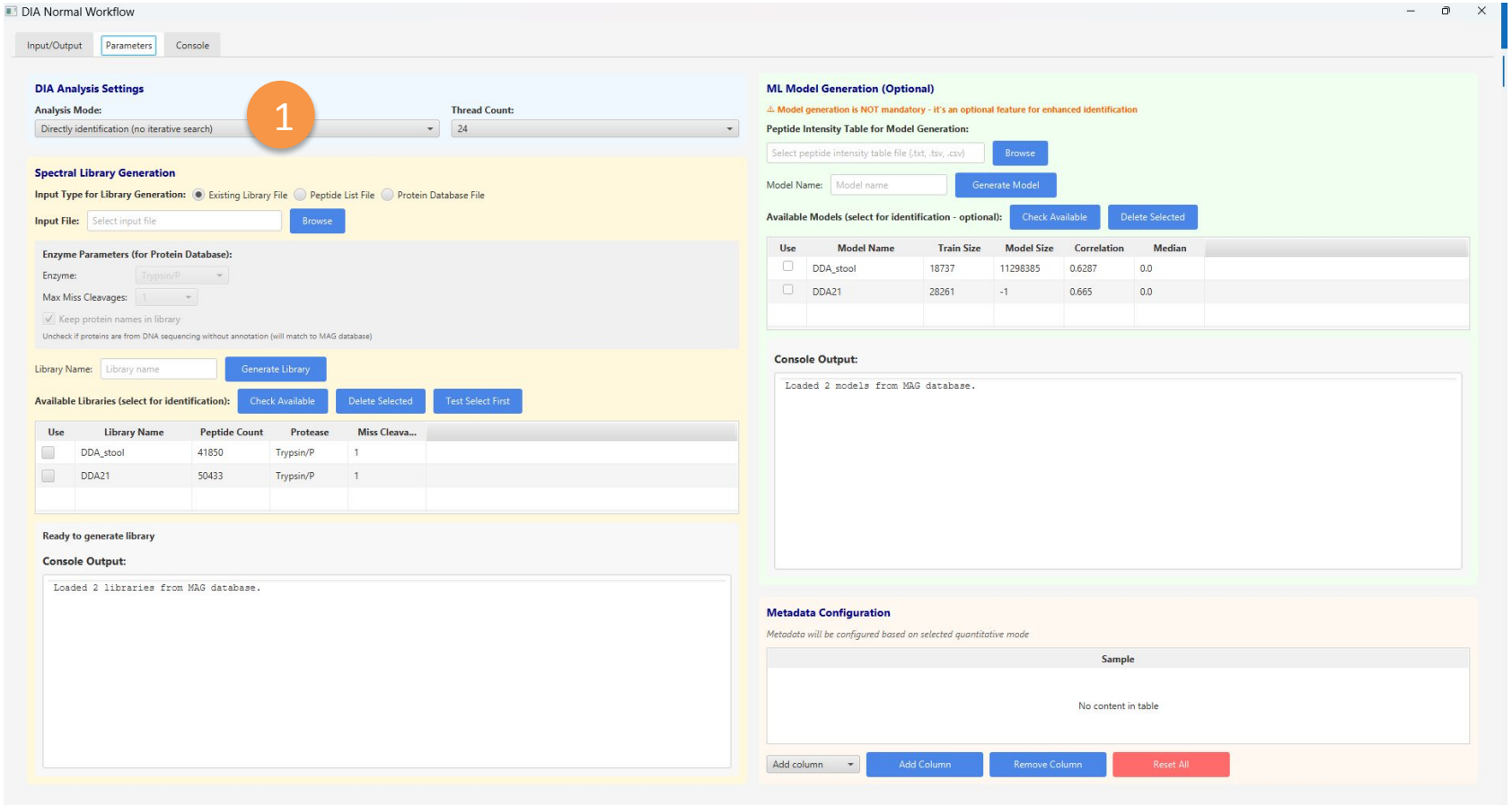


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Running Analyses: DIA Workflows-analysis modes

The normal DIA workflow provides three Analysis Mode:

- Analysis Mode 1: Library Analysis – uses pre-built spectral libraries.
- Analysis Mode 2: Iterative Analysis – creates sample-specific libraries.
- Analysis Mode 3: Self-Model Analysis – ML-enhanced iterative identification.



Step (GUI):

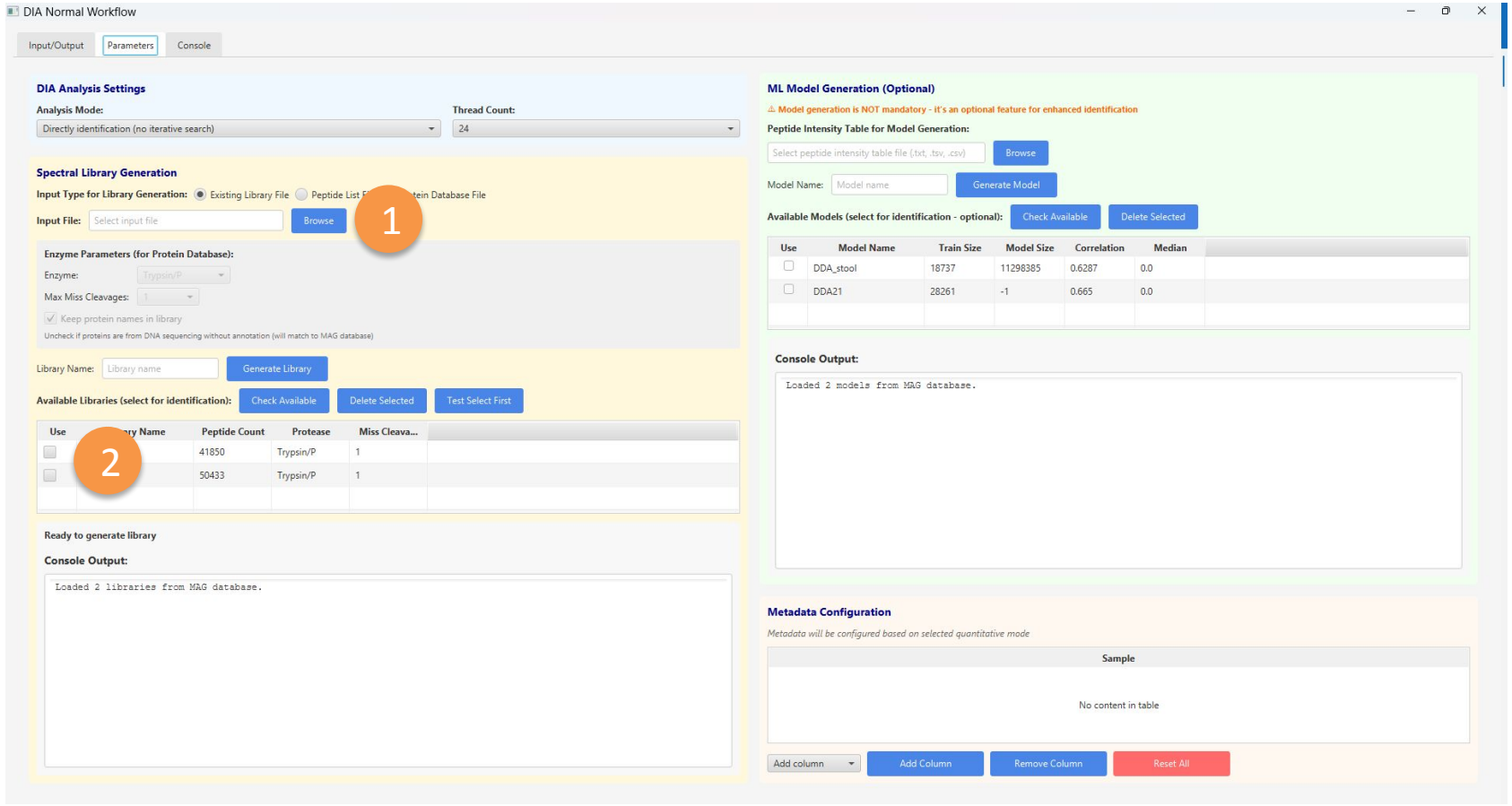
1. Select the Analysis Mode.

Running Analyses: DIA Workflows- library analysis

Library Analysis: fastest method using known spectral libraries.

Step (GUI):

1. Generate new library if no proper one is available; the input file could be a protein database (.fasta) or a peptide list with a column named “Sequence” (.tsv).
2. Choose spectral library.



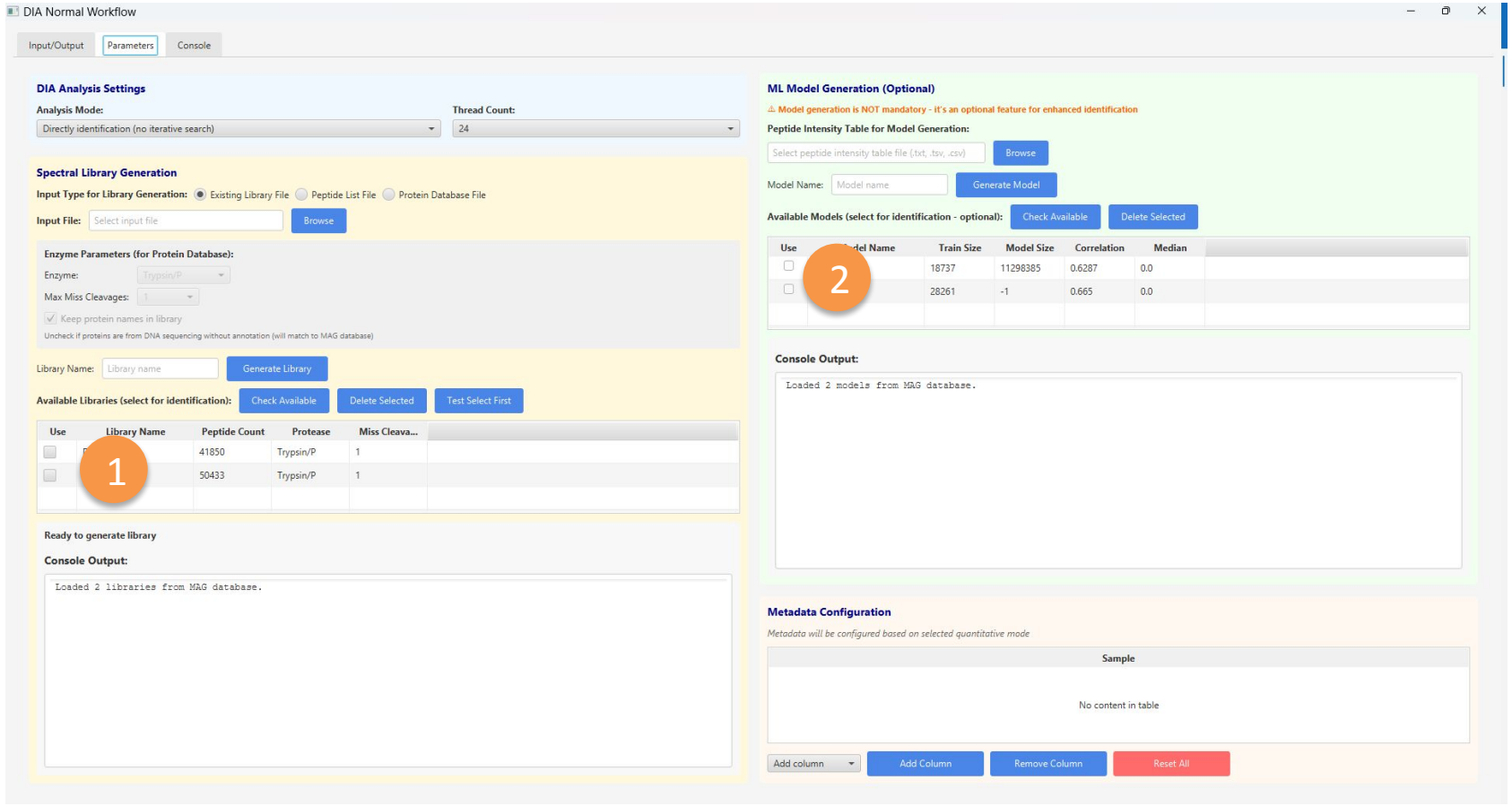
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Running Analyses: DIA Workflows- iterative analysis

Iterative Analysis: builds sample-specific libraries.

Step (GUI):

1. Select broad seed libraries.
2. Optional: Choose ML models to guide rounds. To generate a new model, the input file will be a peptide list with columns named “Sequence” and “Intensity” (.tsv)



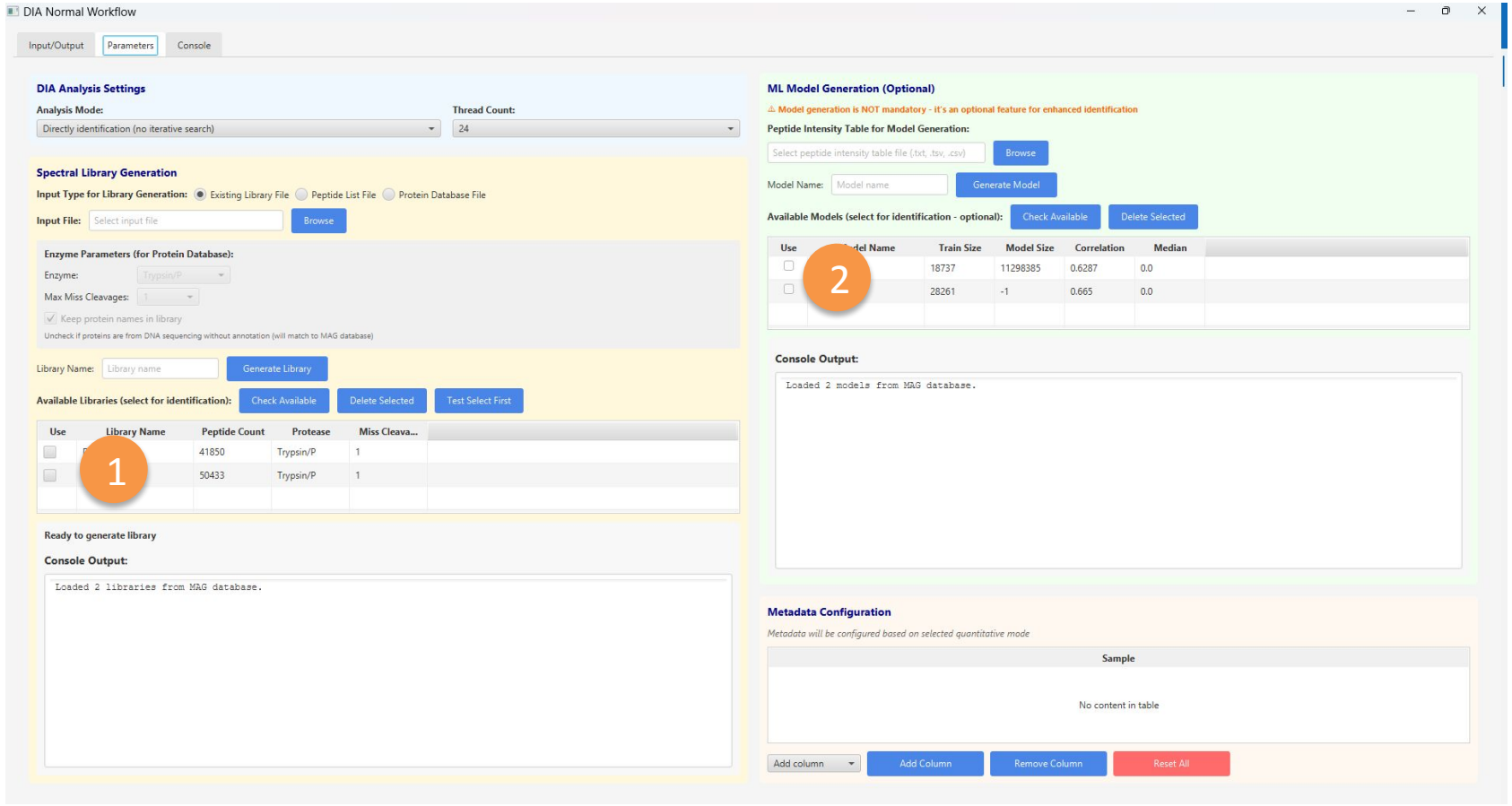
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Running Analyses: DIA Workflows- Self-Model Analysis

Self-Model Analysis: combines iterative search with trained ML models from the dataset itself.

Step (GUI):

1. Select broad seed libraries.
2. Optional: choose ML models to guide rounds.



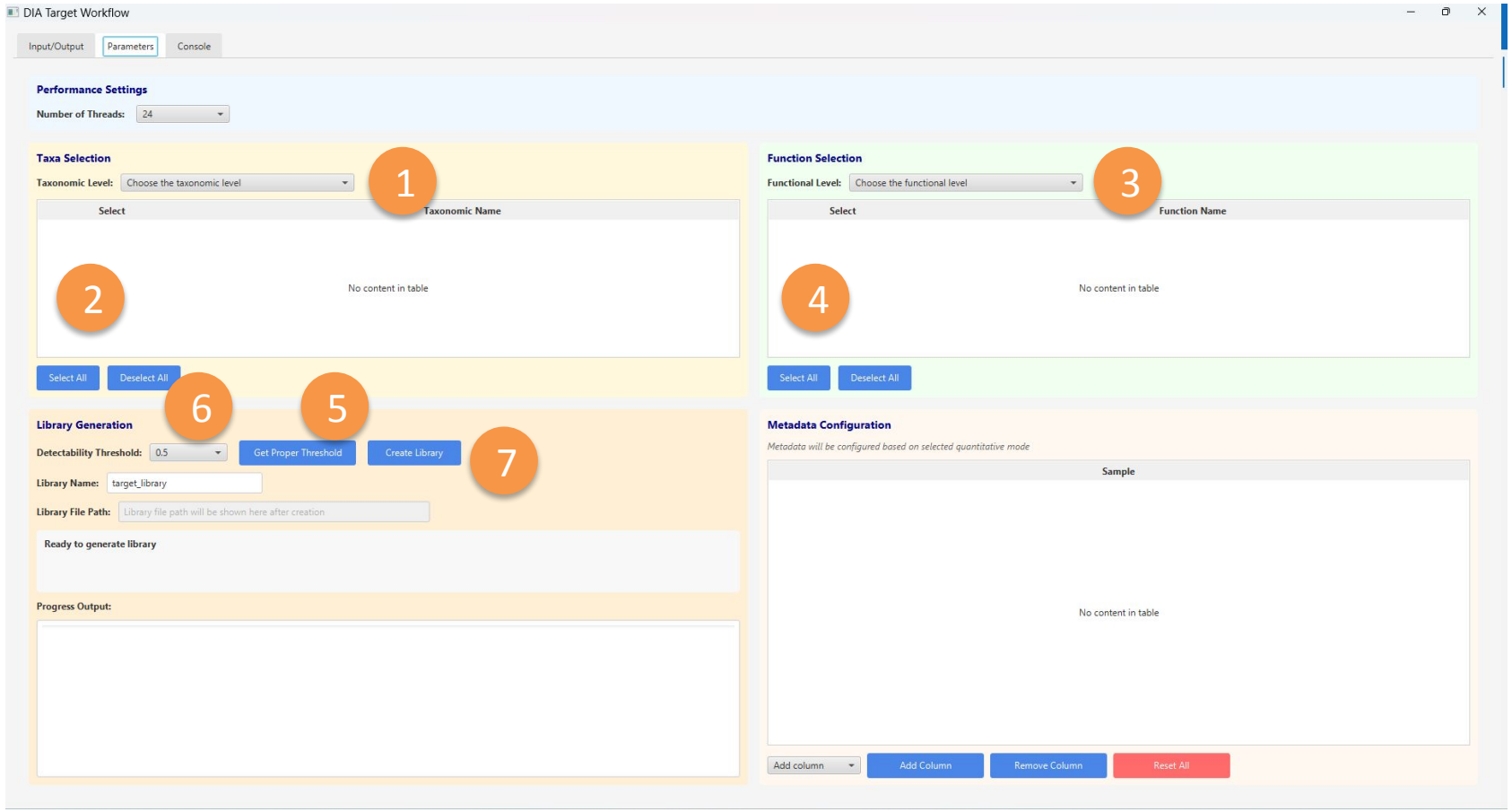
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Running Analyses: DIA Target Workflow

DIA Target Workflow: target proteins by taxonomic group or function.

Step (GUI):

1. Select taxa level.
2. Select taxa.
3. Select function level.
4. Select functions.
5. Calculate the peptide detectability threshold.
6. Choose a proper threshold.
7. Generate the target library.

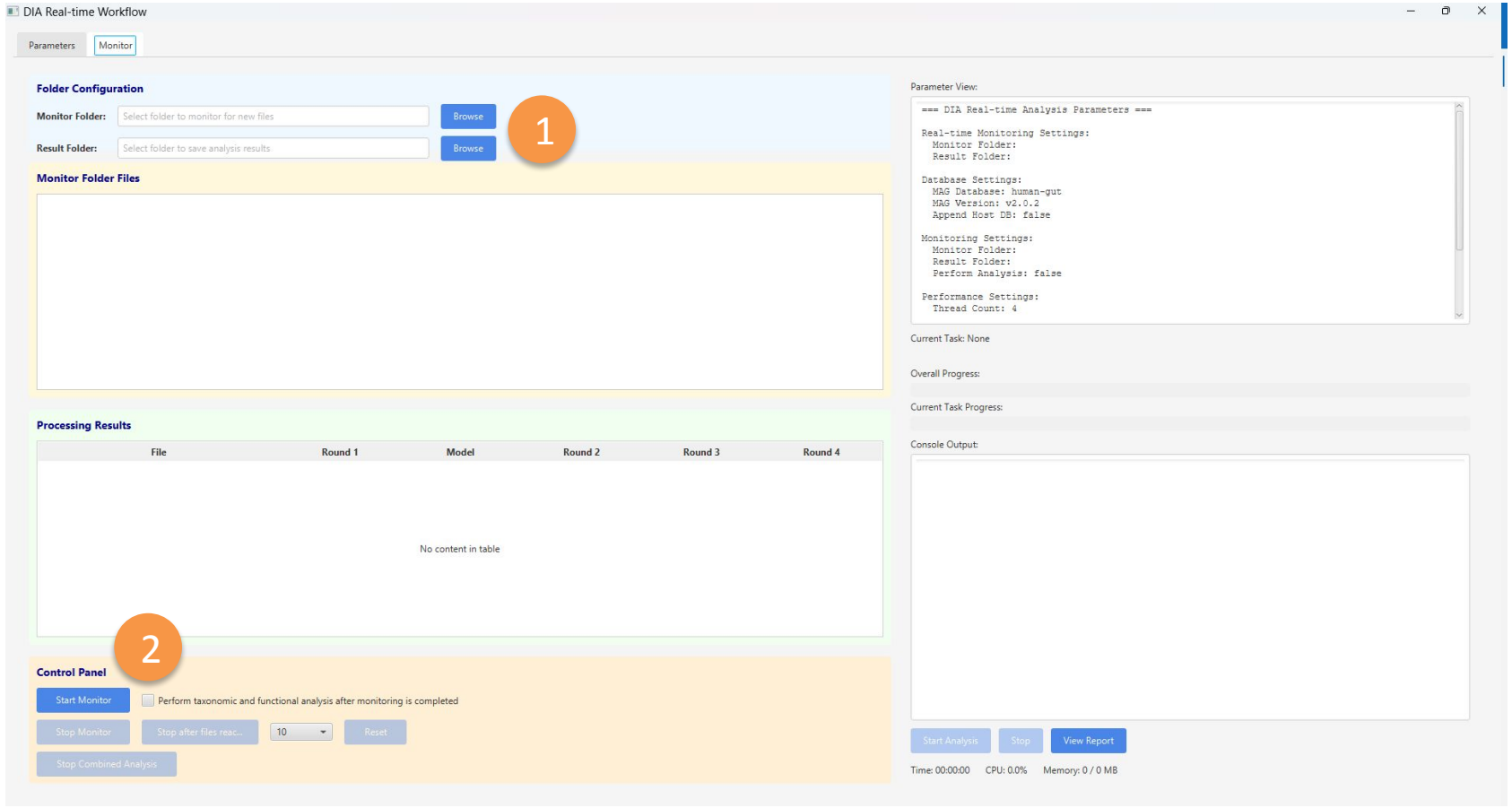


Running Analyses: DIA Real-time Workflow

DIA Real-time Workflow: analyze .raw files live during acquisition. System watches for new files and analyzes them as they complete. Useful for rapid validation and real-time feedback.

Step (GUI):

1. Select monitor folder and result directory.
2. Start the monitor.



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Troubleshooting

Common Issues & Solutions:

- **Ensure all the tools can run successfully in your environment (e.g. if you use the FragPipe workflow, please ensure FragPipe works well in your environment; if you use the DIA workflows, please ensure Dia-NN works well in your environment)**
- Missing tools ⟶ Configure paths under 'Resources' tab
- MAG DB not loading ⟶ Check MAGDB_1_0.json path
- File path errors ⟶ Use absolute paths and verify formats

Support & Version Info

Troubleshooting: See logs/metapilot.log and workflow logs (logging.log and system_output.log).

Version: MetaPilot v1.0.0-beta.2 (Beta)

Contact: MetaPilot Development Team (kai.cheng@quadram.ac.uk)

For updates: https://github.com/cksakura/MetaPilot-Releases/releases